The Evolving Politicization of "Climate" in the United Nations General Assembly Discourse: A Corpus-Based Analysis (1980-2020)

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1. Introduction

Climate stands for a specific location's weather conditions over a long period, typically about 30 years. In other words, it is different from weather, which describes short-term atmospheric conditions. While many people commonly view climate as a stable baseline, it is influenced by several factors, notably natural variations and human activities. Rather than seeing climate merely as what we expect, it is essential to recognize that real climate processes operate through complex interactions which produce unexpected changes over time (Lovejoy, 2013).

The term "climate" has evolved from a neutral scientific concept into a linchpin of global political discourse, particularly within the United Nations General Assembly (UNGA). The latter serves as a microcosm of global climate politics, where member states negotiate responsibilities, vulnerabilities, and solutions. In light of this, this paper investigates the main research question of how do quantitative and qualitative shifts in the usage and linguistic environment of "climate" within the UN General Assembly corpus reflect its increasing politicization and relevance over time and across different actors or representatives of UN member states.

This question is analytically significant for two reasons. First, while climate's political dimensions are widely acknowledged, few studies use corpus linguistics to demonstrate how this politicization manifests linguistically in multilateral forums like the UNGA. Second, by applying tools like collocation analysis and distribution mapping in CQPweb, this paper offers a replicable framework for tracing discursive shifts in diplomatic corpora. From this perspective, this paper addresses the following:

[1] How has the frequency and distribution of "climate" usage evolved in UN General Assembly Discourse over time from 1990-2020 and differ between geopolitical divergences?
[2] What collocational patterns and contextual insights reveal how "climate" has been employed in international discourse?
[3] How did "climate" shift from a general diplomatic term to a political bargaining chip in UNGA discourse (1980–2020)?

Together, these questions trace climate's semantic shift—as evidenced by comparative analysis across decades and geopolitical blocs—through quantitative trends [1], linguistic patterns [2], and their synthesis [3].

"Climate" was selected as a lexical item for its demonstrable relevance in the UNGA corpus and its dual role as both a neutral scientific and a political term. Its high frequency and collocational diversity make it a robust proxy for analyzing how environmental discourse is weaponized, negotiated, and institutionalized in international relations. As the period 1980–2020 has seen critical milestones in this regard—the 1992 UNFCCC**,** 2015 Paris Agreement—making it ideal for tracing how "climate" became entangled with policy negotiations, equity claims, and securitization.

To have a guide to navigate through the rest of this piece of paper, this study is divided into three focused sections that ensure a transparent, data-driven response to the research question while maintaining analytical depth. The first section after the introduction presents how the study was conducted to tackle the research problem and identifies the methodology, justifications for the corpus linguistic tools and procedures employed, and corpus sample of the study. It will address and explain the three mentioned sub-questions in more detail. The Results chapter puts the data at hand by analyzing it and describing the key findings of this study, including visualizations of frequency and distribution trends. Finally, the Discussion chapter organizes, compares, and interprets the significance of the findings through the lens of politicization and provides implications of the study alongside opportunities for future research. Eventually answering this research paper’s concerns in a unified manner.

2. Methodology

CQPweb's flexibility, usability, and powerful analytical capabilities make it an ideal tool for tracing the evolution of the term "climate" in UN General Assembly discourse and for conducting a thorough corpus linguistics analysis within this research framework (Hardie, 2012). Specifically, in order to explore how the discursive patterns of "climate" in the UNGA Debates reflect its growing politicization, this research utilizes essential corpus linguistics methods, which track changes in frequency, collocational occurrences, and distribution across time and geopolitical groups.

To further explore these ideas, this research study uses a mixed-methods approach—a combination of quantitative statistical analysis of language arrangements with qualitative multilateral rhetoric interpretations—with a goal to achieve a detailed understanding of the intersection between language, politics, and global agendas.

2.1. Corpus Description

The UN General Assembly Debates corpus as a whole spans from 1970 to 2020, which amounts to 50 years that witnessed considerable linguistic and cultural evolution. The corpus incorporates yearly delivered statements from UN member state delegates who participate in the general debate to discuss global issues, such as their key responsibilities like working towards human rights, international peace, and sustainable development.

The corpus covers 8,481 of texts, representing a total of 25,972,547 word tokens and 65,454 unique word types as indexed in CQPweb, which contributes to its significance and robustness in corpus-based analysis and research. Moreover, each text is annotated with metadata, including ISO country code, year, and geopolitical grouping, enabling comparative analysis across time periods, regions, and nations.

After initial overall frequency analysis and with the help of the embedded metadata in CQPweb, various subcorpora were sequentially created in alignment with each of the methods and tools under the restricted query section in the CQPweb menu. This enables a more targeted and flexible analysis based on variables such as timeframe (e.g., 2010s) or geopolitical groupings (like Global Nort). For the former, periods were grouped into significant temporal segments or decades, and as for the latter, this paper's author conducted it manually using ISO country codes to reflect the political and linguistic diversity of member states. By leveraging these features, this study ensures a focused, in-depth, and comparative

exploration of the evolving politicization of "climate" in UNGA discourse.

2.2. Analytical Framework

2.2.1. Frequency and Distribution Analysis

To establish the baseline importance of the term "climate" and to create a valuable foundation for understanding its salience in UNGA discourse, frequency analysis was conducted using the query [lemma="climate"%c] in the Standard Query section of CQPweb. This analysis identifies both absolute frequency and relative frequency (instances per million words) for "climate" across the entire corpus.

The distribution analysis shows temporal-related patterns and political-space variations. These insights are crucial for understanding how the politicization of "climate" varies and evolves across actors. Considering this, using the metadata in CQPweb, the temporal distribution of "climate" was analyzed to identify patterns of usage from 1980 to 2020. Similarly, The geopolitical distribution of "climate" was spread by countries (top 20 between 1980-2020) and then was analyzed by grouping countries into the following categories:

- African Group
- Asian and the Pacific Group
- Eastern European Group
- Latin American and Caribbean Group
- Western European and Others Group

The groupings represent the various geopolitical perspectives within the UNGA and the frequency data for "climate" was then analyzed within each group.

This study, therefore, combines temporal and geopolitical distribution analyses to trace both broad trends and specific variations in the politicization of "climate" within UNGA discourse.

2.2.2. Collocation Analysis

The CQPweb platform was used to analyze the semantic environment of "climate" and its politicization in UNGA discourse through conducting collocation analysis. The method reveals which words co-occur frequently with "climate" to show thematic

connections and meaning changes throughout time and between different geopolitical contexts. As a means of illustrating the settings applied in CQPweb, collocations were calculated using a 5-word left/right window with *Log Ratio (filtered)* to identify salient contextual associations. *Word form* was selected over lemma to capture precise terms and avoid conflations that might muddy the analysis. Only collocates with frequency ≥20 (node+collocate) and ≥50 (collocate) were retained, ensuring reliability and statistical significance.

While collocation analysis quantitatively identifies significant word associations (via log ratio), qualitative interpretation is applied to categorize collocates thematically and contextualize them linguistically. The following analytical framework was applied:

- General Collocation Analysis (1980–2020)
- Temporal Analysis (By two decades of 1980s and 2010s)

3. Results

3.1. Frequency and Distribution Analysis

The analysis results reveals as an absolute frequency that the term "climate" appears 12,938 times in the UNGA corpus, with a relative frequency of 498.141 instances per million words. The relative frequency offers a normalized measure and highlights that "climate" is not just frequently mentioned but consistently emphasized across varying corpus sizes. These numbers underscore the prominence of "climate" as a key term in the corpus, mirroring its central role in global political discourse.

As shown in Figure 1, using the metadata in CQPweb accessible through the distribution filter, the temporal distribution of the term "climate" was analyzed to trace its frequency per million words from 1980 to 2020.

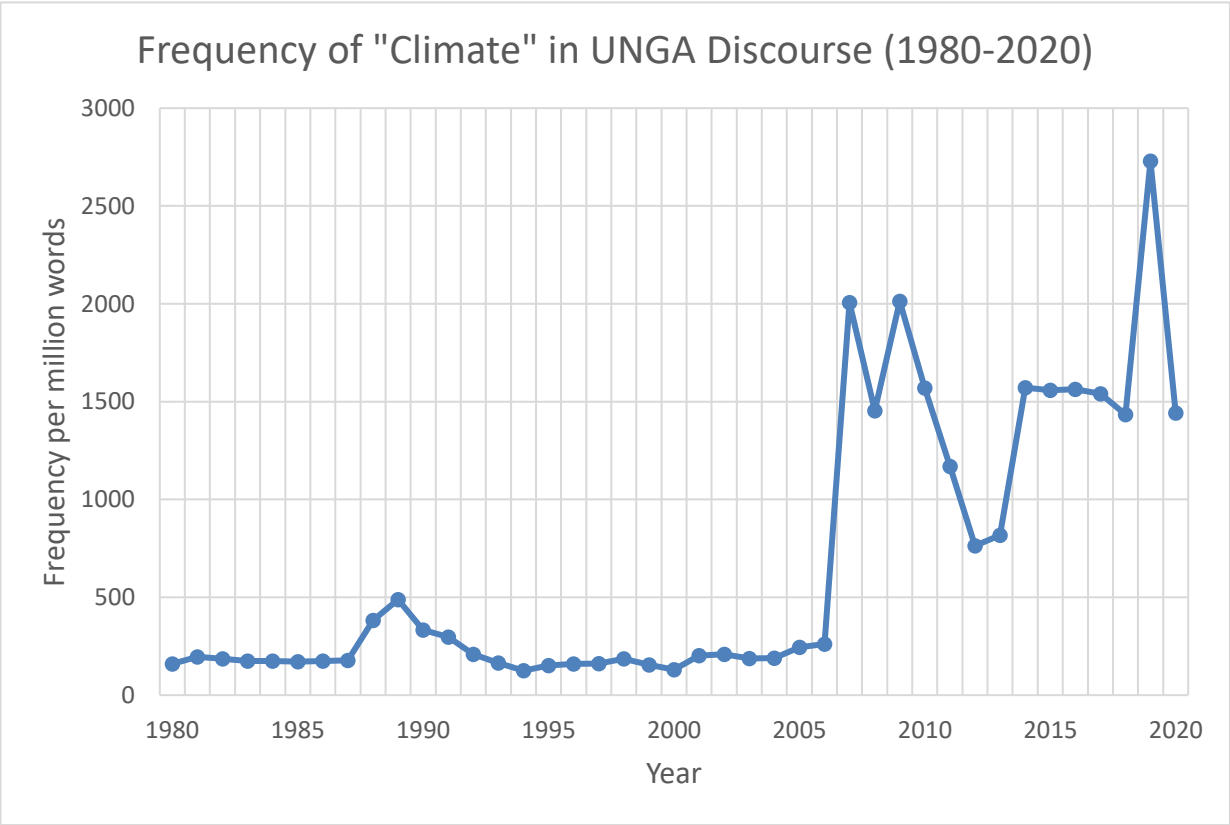

Figure 1

From 1980 to 1987, the word "climate" was used very low and was quite stable,

appearing less than 200 times per million words. This frequency increased in 1988 and 1989, reaching about 383 and 489 times per million words, respectively. In 1990, there was a brief rise to around 334 times per million words, but after that, its use declined and varied between 130 and 300 times per million words from 1990 to 2006. During 2007 to 2010, the usage of "climate" surged significantly, peaking at over 2,000 times per million words by 2009. Following a drop in frequency from 2011 to 2013, it rose again, with peaks of 1,572.01 times in 2014 and an even higher peak in 2019 at 2,729.51 times per million words. In 2020, the frequency reduced again to 1,443.74 times per million words.

The frequency of "climate" was analyzed across countries to identify trends in usage from 1980 to 2020 as demonstrated in Figure 2. The analysis shed focus on the top 20 countries with the highest frequencies, highlighting significant disparities in emphasis on climate-related discourse.

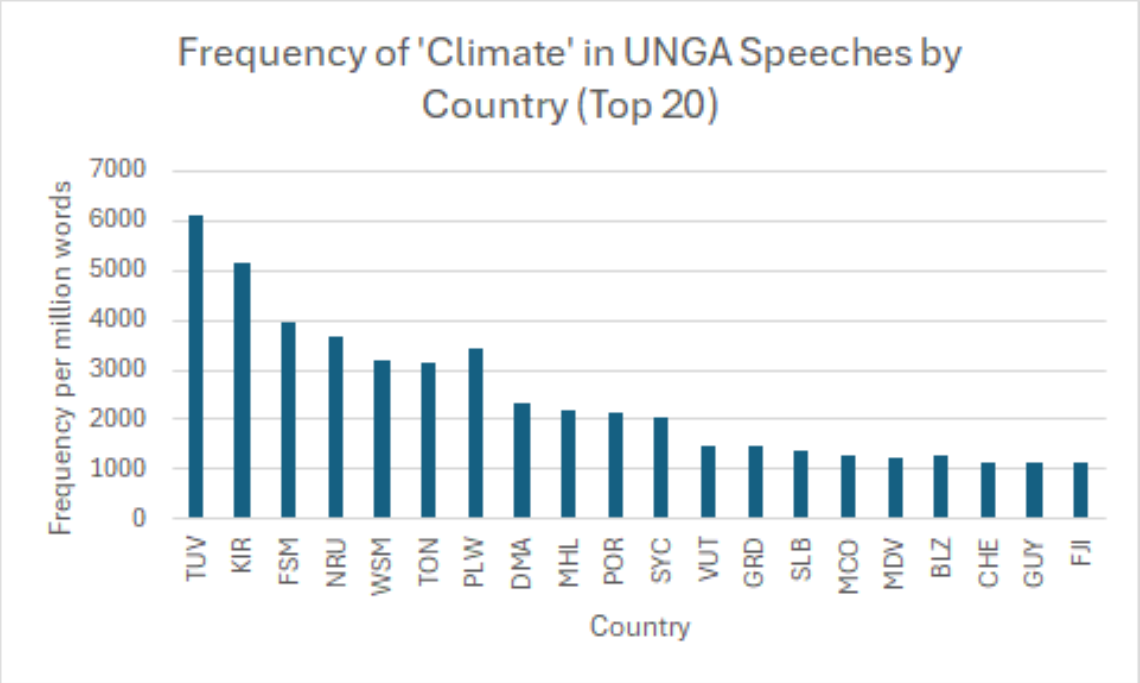

Figure 2

Tuvalu (TUV) is mentioned the most, at 6,097.97 times per million words. Kiribati (KIR) comes next, appearing 5,161.65 times. This high frequency shows that these Small Island Developing States (SIDS) focus heavily on climate issues since they face direct impacts from climate change. Similarly, other Pacific Islands like the Federated States of Micronesia (FSM), Nauru (NRU), Samoa (WSM), Tonga (TON), and Palau (PLW) have more than 3,000

mentions per million words. This further emphasizes how crucial climate issues are to these islands, as they prioritize dealing with the challenges brought by climate change.

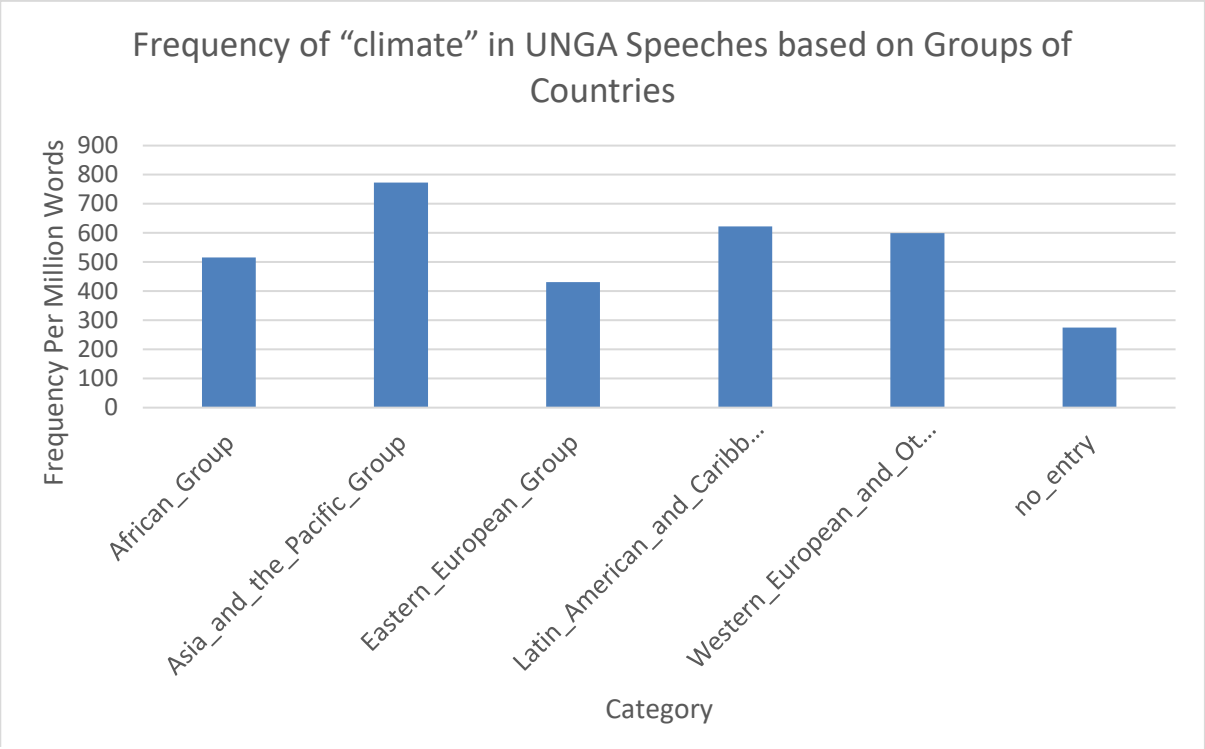

Figure 3

According to Figure 3, the Asian and Pacific Group mentions "climate" the most often, with about 750 mentions per million words. This high number shows they focus a lot on climate issues, especially in Small Island Developing States (SIDS). The Western European and Others Group and the African Group also concentrate a lot on climate, each mentioning it nearly 700 times per million words. The Latin American and Caribbean Group refers to climate less often, with about 500 mentions per million words. The Eastern European Group mentions climate even less, with around 450 mentions per million words, which is the lowest among the major groups. Finally, the "No Entry" category has the least focus on climate, with only about 250 mentions per million words. These findings highlight distinct disparities in climate discourse engagement across geopolitical groups within the UNGA.

3.2. Collocation Analysis

The general collocation analysis of the term "climate" was conducted for the period 1980–2020 (Figure 4), examining frequently co-occurring words to identify patterns of

association within the corpus. The findings establish the most frequent and statistically significant collocations, highlighting recurring themes in climate discourse.

No.	Word	Total no. in this subcorpus	Expected collocate frequency	Observed collocate frequency	In no. of texts	Log Ratio (filtered)
1	change	13,284	80.026	8,221	2371	8.066
2	UNFCCC	88	0.530	46	42	7.498
3	impacts	589	3.548	286	199	7.283
4	induced	119	0.717	43	36	6.545
5	existential	213	1.283	59	58	5.982
6	Paris	1,984	11.952	539	452	5.944
7	mitigating	157	0.946	38	36	5.719
8	adaptation	705	4.247	162	142	5.621
9	mitigation	400	2.410	91	85	5.603
10	adverse	928	5.591	208	177	5.575
11	green	738	4.446	134	116	5.194
12	mitigate	466	2.807	83	80	5.160
13	propitious	166	1.000	29	28	5.126
14	favorable	199	1.199	33	32	5.036
15	effects	3,503	21.103	573	455	5.012
16	resilient	330	1.988	52	41	4.948
17	Panel	705	4.247	110	98	4.931
18	biodiversity	542	3.265	83	76	4.899
19	warming	805	4.850	121	107	4.867
20	conducive	976	5.880	143	131	4.824

Figure 4: Top 20 collocates of "climate" for the period 1980–2020

The observed Kay collocations included words like "change," "impacts," "adaptation," "mitigation," and "effects" which highlighted essential climate discussion aspects. The observed collocate frequency of "change" reached its highest point at 8,221 occurrences together with a significant log ratio of 8.066 which demonstrated its essential role in the discourse. The Institutional collocations "UNFCCC," "Kyoto," "Paris" and "framework" indicate how often international agreements and institutional frameworks appeared in climate-related discussions. The words "adverse," "resilient," "mitigating," "warming" and "disasters" highlight the emphasis on climate change's adverse effects and the strategies for adaptation and mitigation. Terms like "existential," "biodiversity," "green," "binding," and "resilience" signify deeper thematic elements, exploring the broader implications of climate change.

The temporal collocation analysis of "climate" was conducted across two decades, the 1980s and the 2010s, to reveal shifts in associated terms and framing (Figure 5 and 6):

No.	Word	Total no. in this subcorpus	Expected collocate frequency	Observed collocate frequency	In no. of texts	Log Ratio (filtered)
1	change	6,814	100.504	5,470	1507	8.087
2	induced	50	0.738	28	24	6.410
3	adverse	228	3.363	123	108	6.290
4	impacts	422	6.224	220	155	6.185
5	UNFCCC	61	0.900	30	27	6.014
6	Paris	1,184	17.464	539	452	5.803
7	mitigating	67	0.988	30	28	5.759
8	effects	1,014	14.956	401	319	5.449
9	Convention	1,156	17.051	431	383	5.311
10	adaptation	374	5.516	120	105	4.980
11	mitigation	243	3.584	67	62	4.668
12	existential	195	2.876	53	52	4.640
13	mitigate	240	3.540	62	59	4.540
14	Green	557	8.216	132	114	4.375
15	Framework	1,966	28.998	452	405	4.318
16	Fund	730	10.767	150	124	4.111
17	agreement	2,993	44.146	576	471	3.993

Figure 5: Top collocates of "climate" in the 2010s

There are 1,783 different word types in the collocation database for this query (Query "[lemma="climate"%c]", restricted to texts meeting criteria "*year*: *1980* or *1981* or *1982* or *1983* or *1984* or *1985* or *1986* or *1987* or *1988* or *1989*", returned 1,393 matches in 738 different texts) [0.276 seconds – retrieved from cache]

No.	Word	Total no. in this subcorpus	Expected collocate frequency	Observed collocate frequency	In no. of texts	Log Ratio (filtered)
1	conducive	283	0.641	62	53	6.950
2	favorable	185	0.419	32	31	6.527
3	favourable	383	0.867	49	45	6.015
4	improved	361	0.817	38	31	5.697
5	create	1,065	2.411	108	99	5.637
6	improvement	666	1.508	55	48	5.310
7	improving	349	0.790	27	26	5.208
8	creating	498	1.127	37	33	5.145
9	trust	900	2.037	59	56	4.951
10	detente	1,105	2.501	61	47	4.687
11	confidence	1,679	3.800	87	71	4.590
12	creation	932	2.110	47	45	4.549
13	improve	690	1.562	31	31	4.374
14	understanding	1,674	3.789	56	53	3.931

Figure 6: Top collocates of "climate" in the 1980s

The 1980s decade featured essential collocations which included "conducive," "favorable," "create," "trust" and "confidence." The selected words indicate a concentration on mutual understanding and the development of political and international relations. The terms "creation," "improvement" and "better" emphasize the themes of stability and progression in discourse. The log ratio of 3.742 for the collocation "change" indicates that climate change received minimal attention during this time period. The 2010s displayed a significant change in

terminology when compared to the previous decade. The central discussion about climate change emerged through essential words including "change", "impacts", "adaptation", "mitigation""resilience" and "disasters." The term "change" became the most dominant in discourse during this period as indicated by its log ratio of 8.087.

4. Discussion

The corpus-based analysis of the term "climate" in UNGA discourse from 1980–2020 uncovers a clear trajectory of politicization, characterized by changing linguistic patterns, geopolitical disparities, and thematic priorities. The findings are consistent with the study's central research question—how do quantitative and qualitative shifts in the usage of "climate" reflect its growing politicization—while offering a more detailed understanding of the mechanisms of discursive change in multilateral settings. Consequently, the research findings clearly demonstrate that the term "climate" has evolved from its neutral natural/diplomatic origins into a key political tool of international significance.

The temporal distribution analysis shows major frequency increases during important years spanning from 1988 to 1989 and from 2007 to 2010 which demonstrates rising worldwide interest and awareness in climate-related matters. The geopolitical distribution results reveal how climate discourse is used strategically by different groups to advocate for their interests. Small Island Developing States (SIDS) (Tuvalu, Kiribati, etc.) have persistently shown the highest interest in climate discourse (based on Figure 2) because they face direct threats from climate change effects similarly to the Asian and Pacific Group. Meanwhile, regions like Eastern Europe, with lower frequencies, may reflect differing priorities or less direct impacts from climate change.

The semantic evolution of "climate" emerged through collocation analysis, which demonstrated its development from basic diplomatic terminology in the 1980s into action-oriented institutional language during the 2010s, which emphasized "adaptation," "mitigation", and "resilience". The collocations "understanding," "trust," "favorable"," and

“conducive” point toward dialogue encouragement and mutual understanding development but show minimal emphasis on global climate-related concerns. Conversely, the 2010s show a radical change toward urgent and action-oriented language, with collocations like "mitigation," "disasters," and "impacts" dominating the discourse. As well as terms like "Paris," "UNFCCC," and "Framework," which indicate the institutionalization of climate discussions through international agreements. This transition aligns well with the increasing recognition of climate change as a global crisis and the need for coordinated responses and international cooperation. The continuous usage of terms such as "resilience" and "adaptation" enhances the existential threat narrative of climate change, particularly for the above-mentioned vulnerable nations.

This study has limitations despite its strengths, specifically because the corpus ends in 2020, missing out on recent developments in global climate priorities. Future research could expand the corpus to include more recent data that enriches the application of corpus linguistics methods for various latest domains and areas of research. Besides, despite their advantageous outcomes, this study is built solely on frequency, distribution, and collocation analysis, which may not capture the full meaning or contextual nuances present in climate discourse. Additional research into sentiment, rhetorical devices, or pragmatic aspects could give a more wide picture of how “climate” is used and politicized in international forums like the UNGA.

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